

Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework on Elementary Mathematics Learning

A quasi-experimental study on TALPer-supported SRL in fifth grade

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Full Paper • 15 min + 5 min Q&A



A 15-minute storyline: problem -> intervention -> evidence -> implication



TALPer improves mathematics learning, but its strongest lesson is differentiated: students benefit when they regulate, question, monitor, and reflect with AI.

Problem

GenAI may scaffold or create dependence

Intervention

TALPer embedded in SRL 4L

Evidence

Outcome, process, and ENA networks

Takeaway

AI access is not enough

Structured AI feedback can scaffold SRL, but passive use may reduce learner agency



Scaffold

- Real-time feedback
- Step-by-step guidance
- Planning, monitoring, and reflection
- Differentiated support

Dependence

- Passive answer-seeking
- Procedural shortcutting
- Reduced learner agency
- Ability-level differences

Research gap: whether this scaffold-dependence tension differs across learner ability levels remains underexplored.

TALPer is embedded inside the SRL 4L learning cycle

Self-learning -> co-learning -> mutual learning -> teacher-guided consolidation



Original TALPer interface from the source deck



Student asks a mathematical question

TALPer responds through guided reasoning

Interaction logs become classroom evidence

This is the concrete system view behind the SRL 4L integration.

The study follows three evidence layers

Outcome, process, and cognitive network structure



RQ1 · Outcome

Does TALPer + SRL 4L improve learning outcomes compared with SRL 4L alone?

RQ2 · Process

Do high- and low-achieving students differ in SRL strategy use when TALPer is available?

RQ3 · Structure

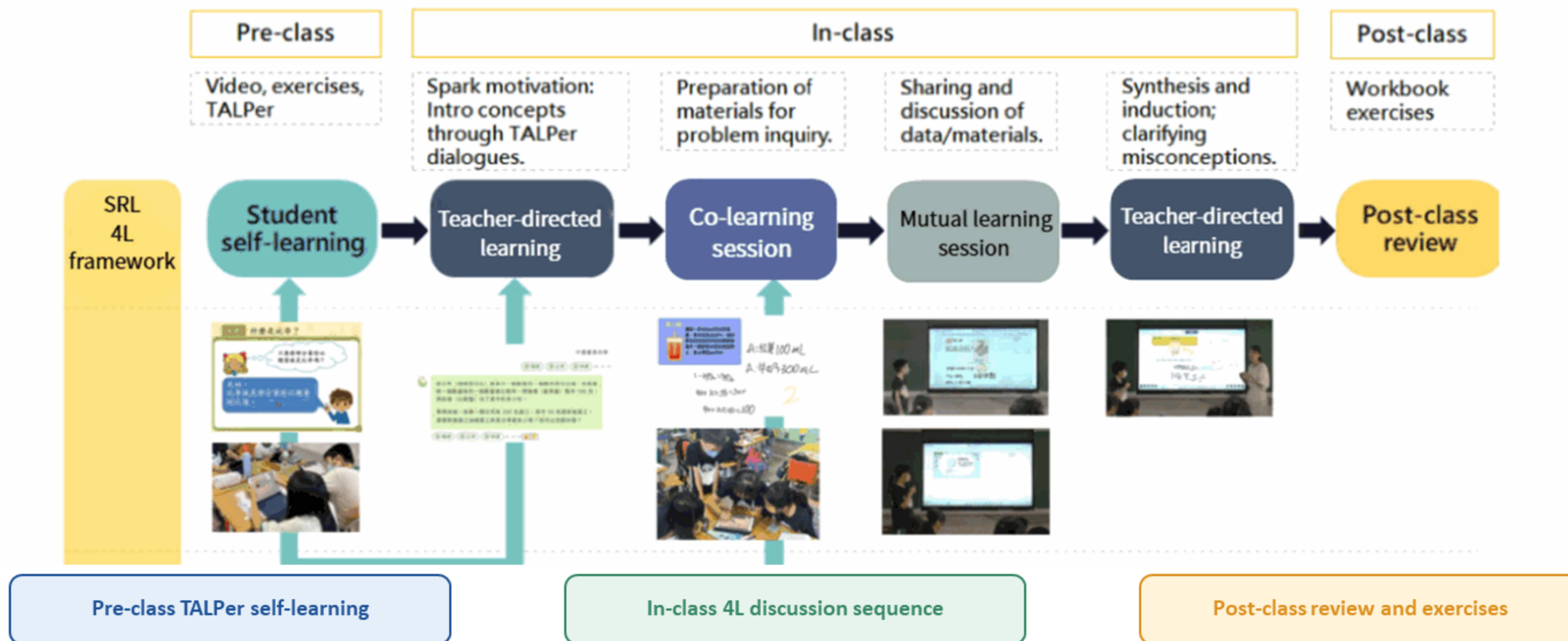
Do their cognitive network structures differ in ENA representations?

The comparison isolates TALPer within the same 4L structure

44 fifth-grade students, 5 lessons on Ratios and Percentages, 2-week intervention



Original SRL 4L teaching process from the source deck



Coding framework connects TALPer dialogue to ENA networks

SRL dialogue coding

Students' TALPer dialogues were coded into metacognitive and cognitive learning behaviors.

Metacognitive

Goal setting

Strategy selection

Monitoring / evaluation

Reflection

Cognitive

Conceptual question

Conceptual explanation

Procedural question

Procedural execution

From codes to ENA

ENA does not only count behaviors. It models how coded behaviors co-occur and form learning-process structures.

1. Segment TALPer dialogue events

2. Apply SRL coding categories

3. Compare co-occurrence networks

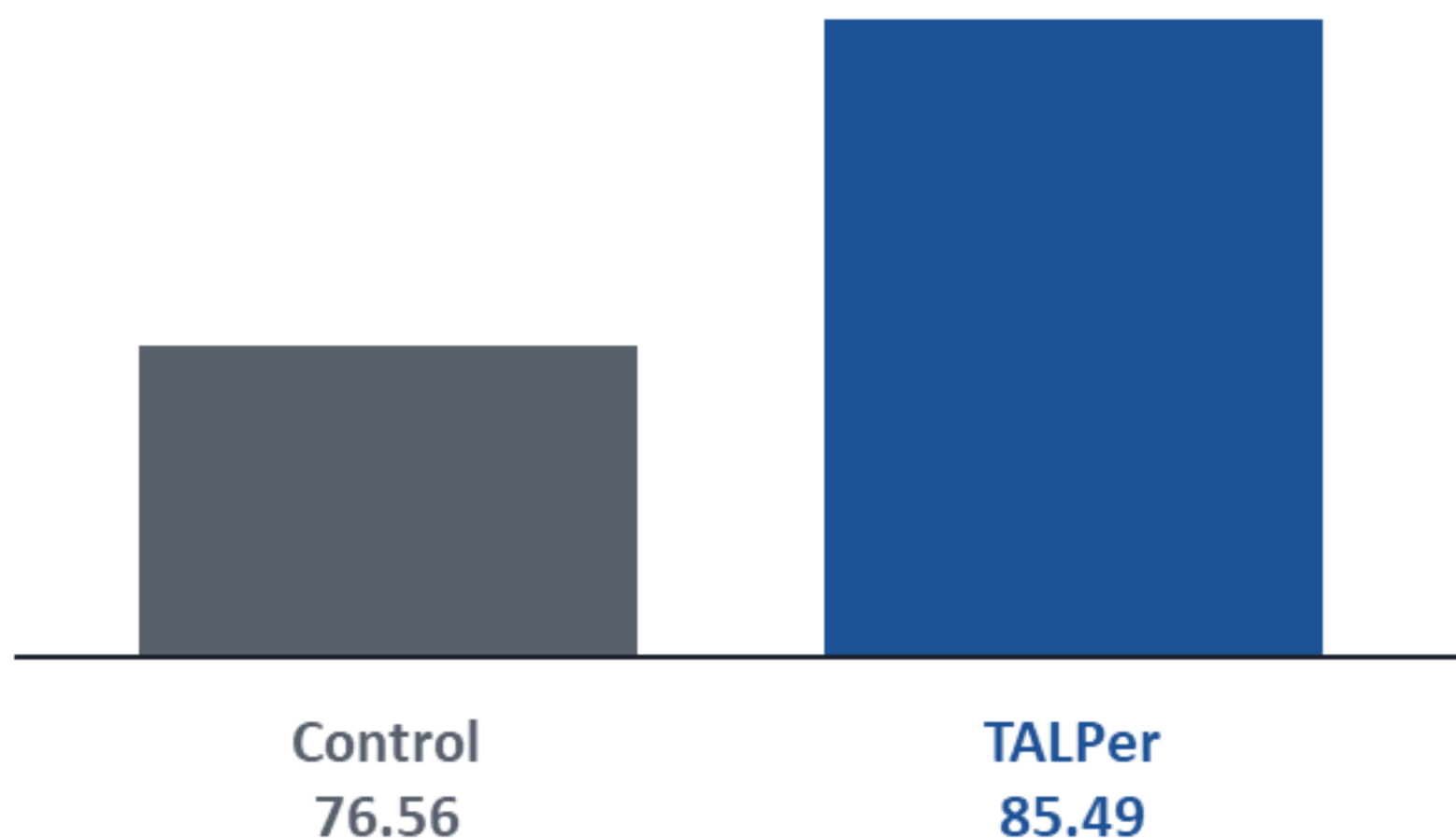
Purpose: show whether students connect planning, inquiry, execution, monitoring, and reflection when learning with TALPer.

TALPer-supported 4L produced higher adjusted post-test scores

After controlling for pre-test performance



Adjusted post-test means



ANCOVA result

$$F(1,41) = 4.71$$

$$p = .036$$

$$\text{eta squared} = .114$$

Takeaway: TALPer adds measurable value beyond SRL 4L instruction alone.

High- and low-achieving students used TALPer differently

The major difference was regulatory strategy, not tool access



High-achieving students

Goal setting
Conceptual questions
Monitoring and reflection
Iterative strategy use

Low-achieving students

Strategy selection
Procedural asking
Answer-seeking flow
Limited reflection

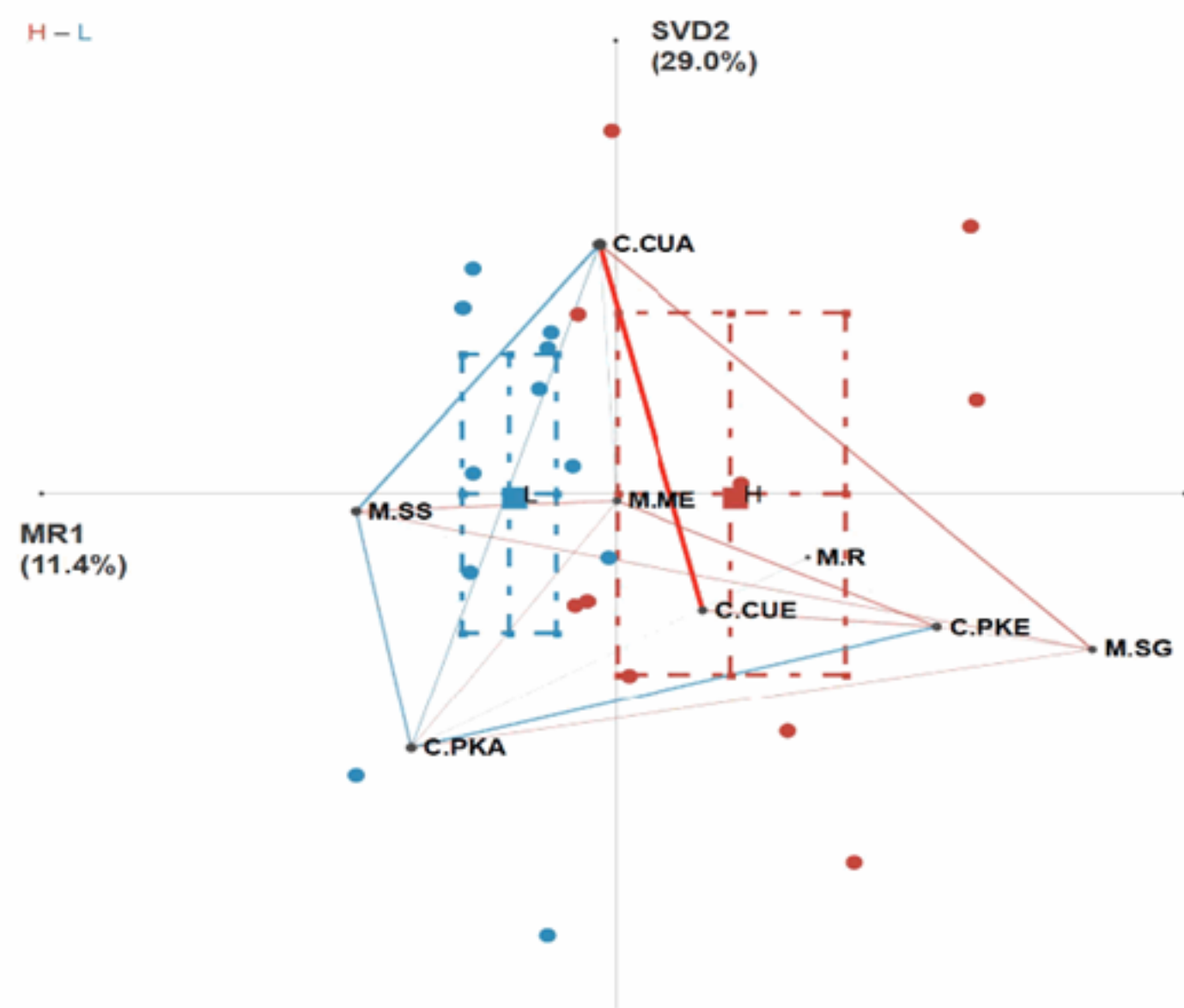
$t(12.07) = 4.05 \cdot p < .001 \cdot \text{Cohen's } d = 1.83$

ENA shows integrated versus fragmented cognitive networks

High-achievers built multidirectional cycles; low-achievers stayed procedural



Original ENA result figure from the source deck



H: integrated, multidirectional regulatory network

L: fragmented, procedural, ask-execute pattern

The simplified editable schematic from v3 has been replaced by the original ENA evidence figure.

Comparison of ENA network structures between high-achieving (H) and low-achieving (L) groups

Availability is not enough; strategic engagement must be scaffolded



TALPer's pedagogical effectiveness cannot be assumed from tool availability alone.

Guided questioning

Model how to ask conceptual questions, not only request answers.

Structured reflection

Use prompts that make students check, explain, and revise.

Dialogue modeling

Show examples of productive TALPer-mediated dialogue.

Thank you. Questions and discussion welcome.



An AI learning companion is not a silver bullet.

Its scaffolding effect depends on learners' regulatory awareness.

Thank you. Questions and discussion welcome.

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